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JAVA Programming Project Presentation

- Project Title: Hotel Room Reservation and Guest Management System
- Course: Java Programming / Mini Project
- Student Name & Roll No: Jarvis Steven J | 25BCS026
- Department: Computer Science
- Mentor / Guide: [Name & Designation]

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Introduction

- This project is a simple JavaFX-based application designed to manage hotel room bookings and guest records efficiently. It provides an easy-to-use interface for reserving rooms, viewing current bookings, and handling guest check-out.
- In many small hotels, room booking is often managed manually, which can lead to problems such as double booking, data confusion, and difficulty in tracking guest details. This project is motivated by the need to create a basic digital solution to simplify these operations.
- The system helps solve real-world issues by ensuring that each room is booked only once at a time, maintaining organized reservation data, and allowing easy check-out of guests. It demonstrates how software can improve accuracy, efficiency, and management in hotel operations.

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Objective

- To design and implement a JavaFX-based application to automate hotel room reservation and guest management tasks.
- To apply core programming concepts such as event handling, data structures (ArrayList), and modular design.
- To develop a system that prevents room conflicts and manages guest check-in and check-out efficiently.
- To demonstrate problem-solving and analytical skills through real-world application development.

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Problem Statement

- In many small hotels, room reservation and guest management are handled manually or with basic systems. This often leads to issues such as double booking of rooms, difficulty in tracking guest details, and lack of proper record management.
- These existing methods are inefficient because they are prone to human errors, time-consuming, and do not provide a clear view of current room availability. Managing check-in and check-out processes can also become confusing without a structured system.
- This project aims to address these challenges by providing a simple JavaFX-based application that automates room reservation, prevents duplicate bookings, and allows easy tracking and check-out of guests, thereby improving efficiency and accuracy.

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System Requirements

Hardware Requirements:

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- Operating System: Windows / Linux
- Programming Language: Java
- IDE: Apache NetBeans
- Framework: JavaFX
- Build Tool: Maven

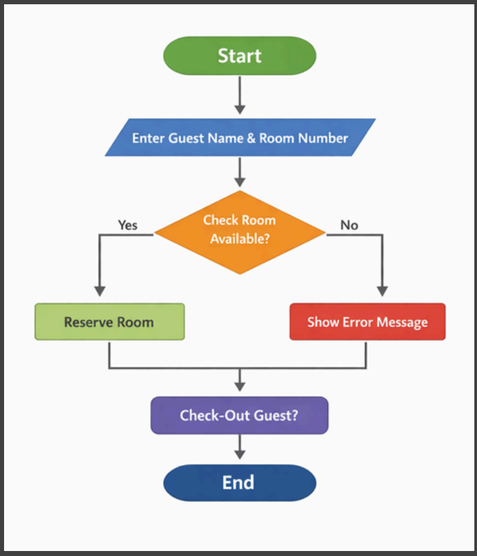
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System Design

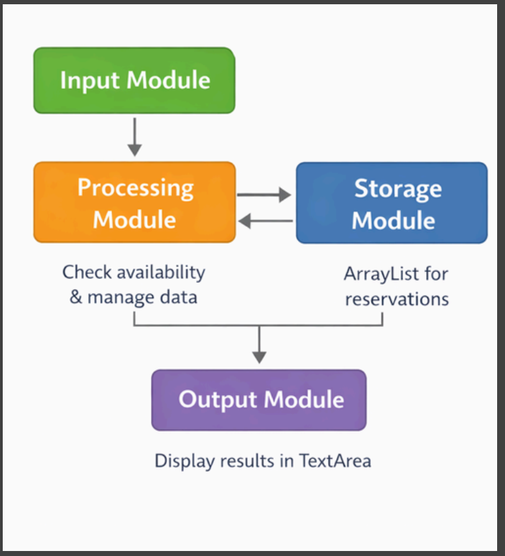
Flowchart (logic flow)



Algorithm outline

1. Start
2. Input guest name and room number
3. Check if room already exists in reservation list
4. If yes → display "Room already bo-bed"
5. Else → add reservation
6. For check-out → search guest name
7. Remove reservation from list
8. Display updated results
9. End

Block diagram (system modules)



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Module Description

Input Module

- Takes user input such as guest name and room number
- Uses TextField components in JavaFX
- Passes input data to the system for processing

Processing Module

- Checks room availability
- Prevents double booking of rooms
- Handles reservation and check-out logic
- Uses ArrayList to manage reservation data

Output Module

- Displays reservation details and system messages
- Uses TextArea to show results to the user
- Provides feedback such as successful booking or errors

File Handling Module

- Currently, the system does not use file storage
- All data is stored temporarily in memory (ArrayList)
- Can be extended in future to save and retrieve data using files or database

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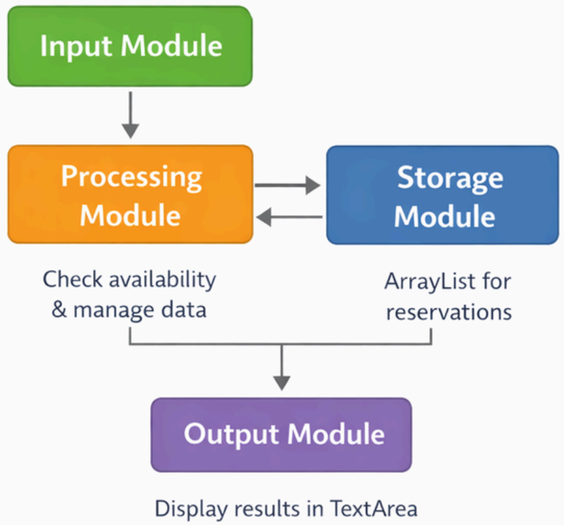
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Algorithm / Flowchart

Algorithm Outline

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2. Input guest name and room number
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Block Diagram (System Modules)



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Implementation

TECHNIQUES USED

Loops and Conditional Statements:

- Used to check room availability and manage reservations using for loops and if-else conditions

Functions (Methods) and Arrays (ArrayList):

- Methods are used to handle button actions, and ArrayList is used to store and manage reservation data dynamically

Event Handling (JavaFX):

- setOnAction() is used to perform operations when buttons are clicked

Data Handling:

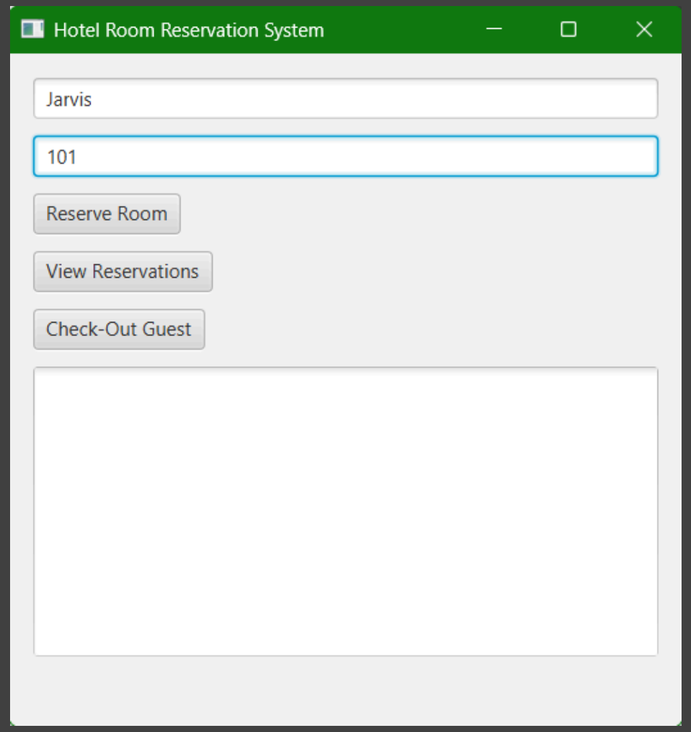
- Data is stored temporarily in memory using ArrayList (no file handling implemented)

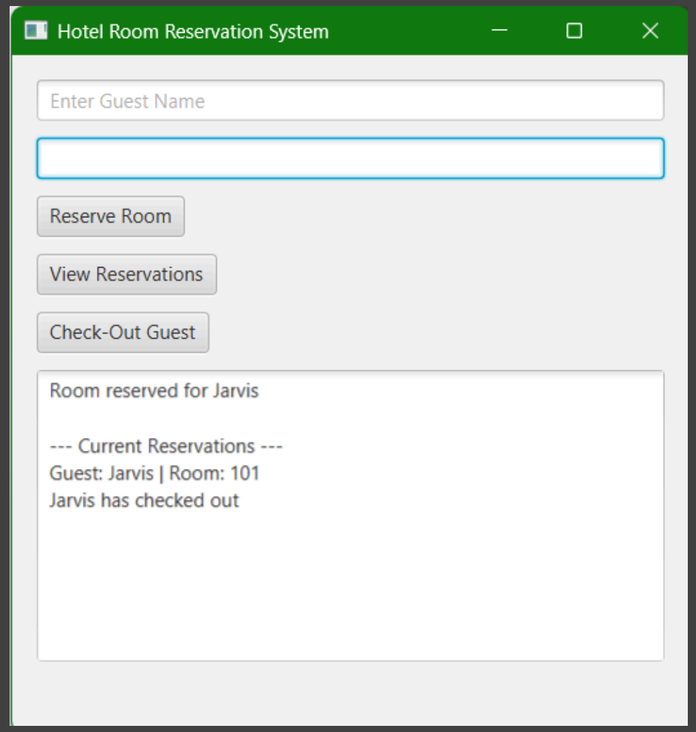
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Output Screens





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Results & Discussion

Summary of Outputs

- The system successfully allows users to reserve rooms, view current reservations, and check out guests
- It displays appropriate messages for successful bookings and errors (e.g., room already booked)

Comparison with Expected Results

- The system performs as expected by preventing double booking of rooms
- All core functionalities such as reservation and check-out are working correctly

Accuracy and Performance Observations

- The system provides accurate results for room availability checks
- Performance is efficient for small-scale usage since data is handled using ArrayList in memory

Challenges Faced and Solutions

- **Challenge:** Preventing multiple bookings for the same room
- **Solution:** Implemented a check using loop and condition to verify room availability

- **Challenge:** Managing reservation data dynamically
- **Solution:** Used ArrayList to store and update data efficiently

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Conclusion & Future Enhancement

Conclusion

- The project successfully achieved its objectives by developing a JavaFX-based hotel reservation system
- It demonstrates key programming concepts such as event handling, loops, conditional logic, and data management using ArrayList
- The system effectively handles room reservation, prevents double booking, and manages guest check-out
- It also improved understanding of GUI development and basic system design

Future Enhancements

- Integrate a database (MySQL) for permanent data storage
- Improve the user interface using advanced JavaFX components (e.g., TableView)
- Add login authentication for secure access
- Extend functionality to handle multiple rooms, categories, and pricing for scalability

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References / Acknowledgement

References

- **Programming in Java** – E. Balagurusamy
- **Official Java Documentation** – Oracle
- **JavaFX Documentation** – OpenJFX
- Online resources and tutorials

Acknowledgment

I would like to thank my mentor and the department for their guidance and support in completing this project.

Thank You